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How Price Dispersion Changes When Upgrades Are Introduced: Theory and Empirical Evidence from the Airline Industry

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Abstract. This paper studies the effect of introducing a new vertical differentiation strategy, paying for an upgrade to a premium product after purchasing the base product, on the price dispersion of the base product arising from existing price discrimination strategies. In particular, we examine how a major U.S. airline's price dispersion in the coach cabin changes after introducing the option to upgrade to a new type of premium economy seating within the coach cabin. We first provide a theoretical analysis that highlights two competing pressures that the new premium economy seating upgrades created on coach class prices. On the one hand, the airline benefits from lowering its prices because by allowing more customers to purchase in the first place, it increases the probability of selling upgrades (admission effect). On the other hand, for some customers, the value of flying with the airline increases because of the upgrade availability, therefore the airline may find it optimal to increase its prices (valuation effect). In the second part of the paper, we conduct an empirical investigation of the impact of upgrade introduction on coach class prices, based on a proprietary transaction-level data set from a major U.S. airline company. The empirical analysis tests the main predictions of our theoretical model and examines further nuances. The results show that the introduction of the premium economy seating upgrades is associated with an increase in the price dispersion and revenues in the coach class, the admission effect is stronger than the valuation effect on the low end of the price distribution, and the opposite is true on the high end of the price distribution. Finally, we discuss implications of our results for firm revenues and consumer welfare.

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1. Introduction

Oftentimes, firms make use of more than one lever of price discrimination to take better advantage of multidimensional customer heterogeneity. The airline industry offers a canonical example of this practice. Although all travelers seek to fulfill the same travel demand, they differ in their preferences for flexible scheduling, comfort, status, and service. Accordingly, airlines commonly engage in vertical differentiation—offering different service levels (e.g., first class versus coach class)—to address heterogeneity in customer valuation for comfort, status, and service. Moreover, within each service class, airlines exploit intertemporal price discrimination—charging higher prices for the same seat when it is booked closer to the flight's departure date—to extract heterogenous customer valuations for flexible scheduling.

Although intertemporal price discrimination has been the main surplus extraction strategy within the coach class, several airlines recently introduced premium economy seats as a new lever of vertical differentiation within the coach class (McCartney 2015). A few examples include United Airlines' "Economy Plus" seating, Delta Airlines' "Economy Comfort" seating, and American Airlines' "Main Cabin Extra" seating. These premium economy seats occupy the first few rows in the coach class and provide features such as greater legroom, extended recline, priority boarding, and complimentary beer and wine. An initial feature of this new option was that it was not offered as a new class but rather as an add-on product. Customers could only purchase the premium economy seats by paying an upgrade fee after they had purchased regular economy seats. Thus, by introducing an upgrade option in addition to its intertemporal price discrimination strategy, the airlines could screen coach cabin passengers in a sequential manner (i.e., first in their valuations for flexible scheduling, then in their valuations for comfort) to extract additional customer surplus that the intertemporal price discrimination lever for the regular product was not effective at appropriating. Although the airline industry is a particularly relevant application, the introduction of upgrades as an additional screening mechanism to intertemporal price discrimination is by no means limited to this industry. For example, in addition to using intertemporal price discrimination based on booking times, many major hotel chains have recently started to offer customers the opportunity to purchase an upgrade to a better room as an add-on product (Stellin 2013), and cruise lines and entertainment venues have followed similar practices.

Motivated by these industry practices, this paper examines the following questions. How does the introduction of vertical upgrades as a new price discrimination lever impact the price dispersion of the firms' regular product? What is the impact of this strategy on the firm and its different customer segments? To answer these questions, we first present a model that captures the main features of the sequential and multidimensional price discrimination context in the airline industry. The model produces several key insights about how the firm's pricing policies for its regular product change in response to the introduction of the upgrade option, and how these changes impact price dispersion, firm revenues, and consumer surplus. Equipped with these insights, we then provide an empirical analysis of how the prices of a major U.S. airline's regular economy seats changed after introducing premium economy seating upgrades. Our focus on the airline industry is motivated by three reasons. First, studying the issue in the airline industry offers us a chance to empirically test theoretical predictions in this domain, taking advantage of a unique data set of individual ticket transactions spanning the timing of the upgrade introduction. Second, the airline industry has been studied extensively as a canonical example of intertemporal price discrimination. Third, this industry has been a leader in implementing revenue management and price discrimination, thus it serves as a great laboratory to test our theoretical predictions.

Our stylized model captures two competing pressures the airline faces in its pricing decisions when facing multidimensionally heterogeneous customers. On the one hand, owing to the sequential nature of the upgrades, the airline benefits from lowering its coach cabin prices after the introduction of the premium seating upgrade option, because allowing more customers to the cabin gives the airline more opportunities to sell upgrades to premium seating (admission effect). On the other hand, customers' value of flying with the airline may increase because of the availability of premium seats, therefore the airline may find it optimal to increase its coach cabin prices (valuation effect). For example, airlines usually offer free upgrades to premium seating for elite members of their frequent-flier programs and may also occasionally upgrade other passengers for free.[1] The free upgrade opportunity may increase customers' willingness to pay when they purchase the tickets. Our model indicates that the result of these competing effects (i.e., admission effect and valuation effect) after the introduction of premium seating upgrades is always associated with an increase in the dispersion of coach cabin prices. The model indicates that whether the prices of tickets bought by business (or leisure) travelers would increase or decrease depends on the relative strength of the admission and valuation effects for that customer segment, paving the way for the empirical analysis to assess the relative strength of these two effects for different customer segments. Furthermore, our model suggests that even though the airline achieves higher revenues and regular prices may increase for some customers, the overall consumer surplus may increase after the introduction of the upgrades.

In the second part of the paper, we offer empirical analyses based on a proprietary data set of individual transactions from a major U.S. airline company spanning a time frame during which the airline introduced the premium seating upgrades in domestic markets.[2] The empirical analyses have two objectives. The first is to test our theoretical predictions regarding the increase in price dispersion within the coach class. In particular, we examine how the introduction of the option to upgrade to premium products (e.g., premium seats in coach) impacts the prices for regular products (e.g., regular seats in coach) in the same market. The second is to shed light on more nuanced empirical questions that the theory model cannot speak to, such as the relative magnitude of price movements for different customer segments and the degree to which price changes impact firm revenues. The data set is uniquely suited for these objectives, because it allows us to identify the price dispersion within a flight instance—a dimension of price variation that is driven by price discrimination based on customer heterogeneity, rather than by factors that may change over time and across flights, such as market conditions, congestion, and service costs.

Our empirical analysis relies on a difference-in-differences strategy that takes advantage of the panel nature of our data and identifies a quasi-experiment for identification. It compares the degree of price dispersion within a flight instance over different instances of the same flight in a "treated" market before and after the market experienced the introduction of the premium seating upgrades versus the degree of price dispersion in flights in a similar "control" market that did not experience this upgrade introduction over the same period. As we discuss in greater detail in Section 5.1, the double differencing alleviates endogeneity and selection concerns that might otherwise lead to spurious correlation.

Our empirical results confirm the theoretical predictions that the price dispersion and revenues increase after the introduction of premium seating upgrades. Furthermore, to understand the mechanism behind this result, we also study how the 10th and 90th percentiles of the price distribution within a flight instance respond to the introduction of premium seating upgrades. We show that the 90th percentile of the price distribution (corresponding to the prices paid by business travelers) increased after the upgrade introduction, whereas the 10th percentile (corresponding to the prices paid by leisure travelers) decreased. Therefore, the empirical results also indicate that the admission effect is to a large extent driving leisure prices down, and the valuation effect is to a large extent driving business prices up. Support for the impact of the admission effect on the low-end prices is further strengthened by an additional empirical finding: the increase in the price dispersion driven by a decline in the low-end prices is more pronounced in markets where the airline faces more competition. Moreover, we show that travelers with higher valuations for late booking not only have a higher willingness to pay for upgrades but also have a higher probability of securing the upgrades with a discount or for free. These findings further lend support to the hypothesis that the valuation effect drives the increase in higher-end prices. Finally, we provide several robustness checks showing the validity of our empirical analyses and generality of our findings.

To our knowledge, our paper is among the first to shed an empirical light on the interaction between different price discrimination levers in a multidimensional price discrimination context. A major strength of the paper is its ability to generate hypotheses relevant to industry practice and to show empirical evidence for them from an industry with sophisticated revenue management techniques, further bolstering our confidence in the applicability of our findings. We hope that our results will be of use to firms across a wide variety of industries as they introduce upgrade options in addition to price-discriminating intertemporally.

2. Literature Review

The main question this paper aims to answer is how the introduction of a secondary price discrimination lever impacts the price dispersion for the firm's main product. This question is closely related to the stream of literature that studies multidimensional screening (e.g., McAfee and McMillan 1988, Armstrong 1996, Rochet and Chone 1998, Armstrong 1999, and Armstrong and Rochet 1999). These papers analyze how a firm that sells multiple product types should design price discrimination mechanisms (e.g., nonlinear pricing) when customers exhibit multidimensional heterogeneity. In these papers, customers make purchasing decisions for different product types simultaneously. However, in our setting, customers first decide whether to purchase the regular product offered by the firm, and only then face the option to upgrade to the premium product. Therefore, the firm screens customers in a sequential manner. In this regard, our application differs from previous multidimensional screening papers in which customers are screened in all dimensions at the same time. Moreover, the focus of our study is on how the firm's screening in the secondary dimension affects the degree of its price dispersion in the primary dimension, a question that has not been the focus of the previous studies. Finally, our empirical results offer a unique contribution to the literature, because to the best of our knowledge, empirical work on multidimensional sequential screening is nonexistent.

Our paper is also related to the operations management literature on how airlines price-discriminate using revenue management techniques (see Talluri and van Ryzin 2005 for a review). Papers pioneered by Gallego and van Ryzin (1994, 1997) have studied dynamic pricing policies for airlines (e.g., Maglaras and Meissner 2006, Besbes and Zeevi 2009, Broder and Rusmevichientong 2012, Chen et al. 2016, and Chen et al. 2018) (see Bitran and Caldentey 2003, and Elmaghraby and Keskinocak 2003 for surveys into this literature). The focus of this literature is on providing efficient algorithms to optimally adjust prices for the flight given the remaining time before the flight. Another stream of theoretical work in operations management studies how dynamic pricing strategies respond to the strategic interactions between firms and consumers (e.g., Su 2007; Aviv and Pazgal 2008; Cho et al. 2009; and Levin et al. 2009, 2010) and between firms (e.g., Liu and Zhang 2013 and Gallego and Hu 2014). In recent work, Stamatopoulos and Tzamos (2018) also study how dynamic pricing in airlines may interact with the airline's endogenous choice of legroom. Chun et al. (2017) and Chun and Ovchinnikov (2017) study the interaction between airline revenue management and the premium-status loyalty program. In this paper, we study how the airline's price discrimination policy responds to the offering of a new premium economy seating.

The operations management literature also features recent research that studies upgrades and upsells (e.g., Aydin and Ziya 2008, Gallego and Stefanescu 2009, Cui et al. 2018, Çakanyıldırım et al. 2017, and Yılmaz et al. 2017) and cross-selling (e.g., Cui and Shin 2018). These papers assume that upgrades are available and focus on when offering customers free or paid upgrades will benefit the firm and how upgrades should be priced. Differently, this paper studies how the firm's upgrade policy interacts with its primary price discrimination policy. We focus on how the offering of upgradable premium products affects the degree of price dispersion observed for regular products.

The airline industry has been of interest to many empirical researchers owing to its importance in the economy and its multifaceted nature. Researchers in operations management have empirically examined the impact of baggage fees (Nicolae et al. 2017) and flight schedules (Deshpande and Arıkan 2012) on on-time departures, how airlines use different robust scheduling practices to maximize operational efficiencies (Atkinson et al. 2016) and how operational efficiencies impact contemporaneous stock returns (Ramdas et al. 2013), whether consumers are forward-looking (Li et al. 2014), and how domestic airline alliances impact market structure (Li and Netessine 2011).

However, the most active area of empirical research in the airline industry has been the exploration of different factors that influence price dispersion. Among these factors, the impact of competition received the most attention and debate. Early studies that rely on cross-sectional analysis, such as Borenstein and Rose (1994) and Stavins (2001), find that more competition is associated with higher levels of price dispersion. However, more recent studies that use a panel analysis approach, such as Gerardi and Shapiro (2009) and Gaggero and Piga (2011), find that a higher degree of competitiveness leads to a decrease in price dispersion. Finally, Dai et al. (2014) argue that the effect of competition on price dispersion is nonmonotonic. More recently, researchers have explored other factors, such as the effect of price formats (Chellappa et al. 2011), dispersion across online travel agencies (Clemons et al. 2002), business cycles (Cornia et al. 2012), and the interaction between time left to departure and market conditions (Mantin and Koo 2009).

Our empirical investigation also examines changes in the degree of price dispersion. One of the main advantages of our empirical analysis is that our data allow us to identify the distribution of coach class prices in each flight instance. In contrast, previous studies of price dispersion in the airline industry have relied on the DB1B database, which is a 10% random sample of domestic tickets drawn from a given market across a quarter of a year.[3] Not only does the DB1B database not distinguish the date of departure, it also conflates price dispersion arising from three different sources: price variation across flights departing on different dates, across flights departing at different times of the day, and within a given flight instance, across passengers. Previous literature debates the attribution of this composite variation to price discrimination (see, for example, Hayes and Ross 1998 and Cornia et al. 2012). Because of the disaggregate nature of our data, we uniquely contribute to the empirical literature on price dispersion by studying the impact of the introduction of premium economy seating upgrades on the coach cabin price dispersion within a flight instance that precisely results from intertemporal price discrimination. We also explore whether this impact varies by the competitiveness of the market.

3. Model

We consider a firm that can either sell regular products only or sell regular products and give customers the opportunity to upgrade to premium products by paying an upgrade fee after purchasing regular products.[4] Compared with regular products, premium products have additional features that enhance customer utility in addition to the utility generated by the common features of both regular and premium products. For example, although both regular economy seats and premium economy seats fulfill the transportation need of customers, the additional features of premium economy seats increase customer utility by enhancing comfort and convenience.

Customers are heterogeneous in their valuations for regular products, as well as their probabilities of upgrading to premium products. Customers' valuations for regular products, v, are uniformly distributed between 0 and 1, and the total customer mass is normalized to be 1. As is typical in the airline industry, leisure travelers have lower willingness to pay and book tickets earlier in advance, whereas business travelers have higher willingness to pay and book tickets closer to the travel date. On the basis of this feature, airlines charge lower prices early in the booking period to induce leisure travelers to purchase and charge higher prices later to business travelers to extract more surplus from them. For example, analyzing more than 12,000 price observations, Etzioni et al. (2003) find that airlines usually significantly increase their prices 2 weeks before the departure date. To capture these regularities, we assume that customers arrive to the market in the increasing sequence of their valuations v. Correspondingly, the firm first sells regular products at price p_1 when lower-valuation customers arrive and then increases the price to p_2 to sell to higher-valuation customers. Thus, customers with valuations $p_1 \le v < p_2$ purchase at price p_1, those with valuations $v \ge p_2$ purchase at price p_2, and those with valuations $v < p_1$ do not purchase.[5] These model features mimic airlines' practice of price-discriminating customers according to their booking times.

First, consider the case in which the firm only sells regular products. The firm's goal is to choose the optimal prices $p^*_{1,r}$ and $p^*_{2,r}$ (where the subscript of "r" indicates that the firm only sells regular products) so that the revenue from selling regular products is maximized. Given the model setup, the firm sells $1 - p_2$ amount of regular products at price p_2 and $p_2 - p_1$

amount of regular products at price p_1. The firm's pricing problem is thus formulated as follows:

$$\begin{aligned} \max \quad & \Pi_r(p_1, p_2) = p_2(1-p_2) + p_1(p_2 - p_1) \\ \text{s.t.} \quad & 0 \le p_1 \le p_2 \le 1. \end{aligned} \tag{1}$$

The firm's optimal revenue is $\Pi_r^* = \Pi(p_{1,r}^*, p_{2,r}^*)$.

Next, consider the case in which the firm sells premium products as well. The firm's upgrade price is $p_u > 0$. We assume $p_u \le \frac{3}{4}$ to ensure that the revenue function is jointly concave in p_1 and p_2. Given that the maximum v is 1, this assumption is not restrictive, especially because airlines' premium economy seating upgrade price tends to be only a small fraction of the economy seating prices. In our basic model, the upgrade price is exogenously determined, to be consistent with the practice of the airline we worked with as well as our empirical analysis.[6] In Section 4.3, we extend our basic model to study an alternative model in which the upgrade price is jointly optimized with product prices. Our hypotheses are unaffected by this extension.

The model also allows for the fact that customers' valuations from purchasing regular products may increase when the opportunity to upgrade to premium products is offered. This increase may be driven by the fact that airlines usually offer free upgrades to elite members of its frequent-flier program and may also occasionally upgrade other passengers for free. To capture this effect, we assume that when the firm sells premium products through upgrades, customers' valuations become $\hat{v} = (1+w)v$, where v is the original valuation for regular products. The multiplier $w \ge 0$ measures the magnitude of upgrade-driven valuation enhancement. In this formulation, the customers who have higher valuations for the regular product experience a higher valuation enhancement when the option to upgrade to premium products is offered. This formulation choice reflects the observation that higher-valuation customers are more likely to be frequent fliers and hence are more likely to receive free upgrades. We provide the empirical evidence in Table A.2 of Online Appendix A and discuss it further in Section 5.4.[7] Additionally, our results are robust even if the upgrade-driven valuation enhancement takes the additive form instead of the multiplicative form.

To calculate the expected revenue from selling paid upgrades, we need to consider the probability that each customer pays to upgrade to a premium product. Let $u + \rho(v - \frac{1}{2})$ denote the probability that a customer with original valuation v would like a premium product but does not receive the upgrade for free, where $u \in (0,1)$ is the probability that the average customer (i.e., the customer whose original valuation is $v = \frac{1}{2}$) purchases an upgrade. Note that we can equivalently write the upgrade probability as $u + \rho(\frac{\hat{v}}{1+w} - \frac{1}{2})$. Customers' probability of purchasing an upgrade is a linear function of their valuations for regular products, where the slope ρ can be either positive or negative.[8] If $\rho \ge 0$, the firm is more likely to earn upgrade revenues from customers with higher valuations for regular products. If $\rho < 0$, the firm is more likely to earn upgrade revenues from customers with lower valuations for regular products. In Table A.1 of Online Appendix A, we provide the empirical evidence that $\rho \ge 0$ is more likely to occur. We assume that $0 \le u + \frac{\rho}{2} \le 1$ and $0 \le u - \frac{\rho}{2} \le 1$ to ensure that the upgrade probability is between 0 and 1.

Given these primitives, the firm chooses the optimal discriminatory regular product prices, p_1^* and p_2^*, so that the total revenue from selling both types of products is maximized. The firm's optimization problem is formulated as follows:

$$\begin{aligned} \max \ \Pi(p_1, p_2) &= p_2 \frac{1+w-p_2}{1+w} + p_1 \frac{p_2 - p_1}{1+w} \\ &\quad + p_u \int_{p_1}^{1+w} \left[u + \rho\left(\frac{\hat{v}}{1+w} - \frac{1}{2}\right)\right] \frac{1}{1+w} \, d\hat{v} \\ &= \frac{1}{1+w}[p_2(1+w-p_2) + p_1(p_2 - p_1)] \\ &\quad + p_u \int_{\frac{p_1}{1+w}}^{1} \left[u + \rho\left(v - \frac{1}{2}\right)\right] dv \qquad (2) \\ \text{s.t.} \quad & 0 \le p_1 \le p_2 \le 1 + w. \end{aligned}$$

The firm's optimal total revenue is $\Pi^* = \Pi(p_1^*, p_2^*)$. Note that $\hat{v}$ is uniformly distributed over $[0, 1+w]$, and its probability density is $\frac{1}{1+w}$. The firm sells $\frac{1+w-p_2}{1+w}$ amount of regular products at price p_2 and $\frac{p_2-p_1}{1+w}$ amount of regular products at price p_1. $\int_{p_1}^{1+w}[u + \rho(\frac{\hat{v}}{1+w} - \frac{1}{2})]\frac{1}{1+w} d\hat{v}$, or $\int_{\frac{p_1}{1+w}}^{1}[u + \rho(v - \frac{1}{2})]dv$, is the expected number of customers that purchase upgrades. Thus, the firm's revenue function is given by (2). Finally, we note that our basic model is uncapacitated. It is relatively rare for airlines to have a flight with 100% load factor constantly (if tickets are constantly sold out for a flight, the airline would have either moved to a larger aircraft or increased the price). We also provide the analysis for a capacitated model in Online Appendix B.

4. Effects of Upgrades

In this section, we analyze the model we developed in Section 3. On the basis of the results we obtain, we also develop hypotheses to test in our empirical analysis. To study the effect of offering the option to upgrade to premium products on the firm's price discrimination strategy for its regular products and generate insights regarding firm revenue and consumer surplus, we need to compare the case with premium products with the case without. We first study the response of individual prices, p_1 and p_2, to the introduction of the upgrade option. This analysis will lead to an understanding of the change in price dispersion of regular products as a response to the introduction of the upgrade option. Then, we investigate the implications for firm revenue and consumer surplus. All proofs of propositions are provided in Online Appendix C.

4.1. Effects on Price Dispersion and Main Empirical Hypotheses

We first study the change of individual prices due to the upgrade introduction. Proposition 1 states that both p_1 and p_2 increase after the firm starts offering the option to upgrade to premium products if the upgrade-driven valuation enhancement is significant (i.e., w is large) but decrease if w is small enough.

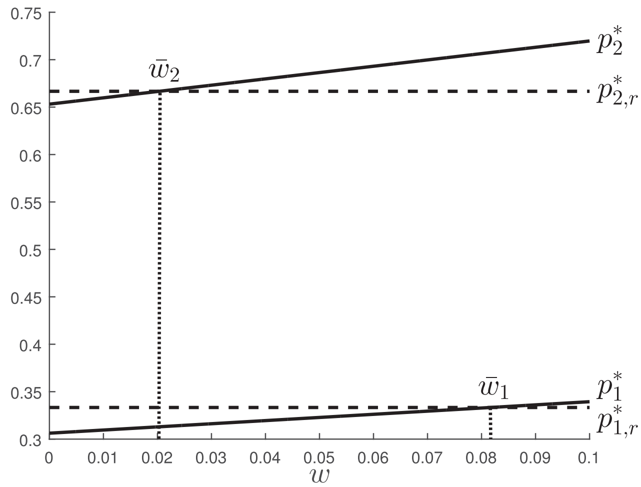
Proposition 1. (i) *There exists a threshold $\bar{w}_1$ such that $p_1^* \geq p_{1,r}^*$ if and only if $w \geq \bar{w}_1$.* (ii) *There exists a threshold $\bar{w}_2$ such that $p_2^* \geq p_{2,r}^*$ if and only if $w \geq \bar{w}_2$.*

The introduction of premium product upgrades changes the firm's optimal pricing of regular products in two ways. On the one hand, because of the sequential nature of the upgrades, the firm would like to decrease the prices to allow more customers to purchase regular products because some of these customers will also purchase upgrades to premium products. We refer to this as the *admission effect*. On the other hand, because customers' valuations for regular products increase with the option to upgrade to premium products, the firm may find it optimal to increase the prices of regular products. We refer to this as the *valuation effect*. For both p_1 and p_2, when the valuation effect dominates the admission effect, the price increases after the upgrade introduction, and vice versa. Figure 1 illustrates how the optimal prices change with w. In this example, $p_1^* \geq p_{1,r}^*$ if $w \geq \bar{w}_1 = 0.0816$, and $p_2^* \geq p_{2,r}^*$ if $w \geq \bar{w}_2 = 0.0203$. If $w < 0.0203$, both p_1 and p_2 decrease after the upgrade introduction; if $0.0203 \leq w < 0.0816$, p_1 decreases, whereas p_2 increases after the upgrade introduction; if $w \geq 0.0816$, both prices increase after the upgrade introduction. Notice that in Figure 1, $\bar{w}_1 > \bar{w}_2$. We next show that this is always the case.

Proposition 2. *After the upgrade introduction, p_2 is more likely to increase than p_1 (i.e., $\bar{w}_1 > \bar{w}_2$).*

Proposition 2 states that the threshold w above which the optimal price increases after the upgrade introduction is higher for p_1 than for p_2. This indicates that p_2 (the price charged to business travelers) increases after the upgrade introduction in more cases compared with p_1 (the price charged to leisure travelers). Thus, the admission effect is dominated by the valuation effect in more cases for p_2 than for p_1. Because the way that the upgrade introduction affects individual price levels depends on the actual value of w, whether each price increases or decreases after the upgrade introduction is an empirical question. In Section 5, we empirically test whether p_1 is driven more by the admission effect or the valuation effect, and whether p_2 is driven more by the admission effect or the valuation effect, by testing between three empirical possibilities: (1) p_2 decreases, p_1 decreases more; (2) p_1 decreases, p_2 increases; (3) p_1 increases, p_2 increases more. We refer to the collection of these possibilities as Hypothesis 1a–1c.

Figure 1. Optimal Prices as Functions of w



Note. $p_u = 0.2$, $u = 0.3$, $\rho = 0.5$; in this example, $\bar{w}_1 = 0.0816$, $\bar{w}_2 = 0.0203$.

Hypothesis 1a. *High-percentile prices decrease, and low-percentile prices decrease more.*

Hypothesis 1b. *Low-percentile prices decrease, and high-percentile prices increase.*

Hypothesis 1c. *Low-percentile prices increase, and high-percentile prices increase more.*

Proposition 3. *p_2 increases more than p_1 after the upgrade introduction (i.e., $p_1^* - p_{1,r}^* < p_2^* - p_{2,r}^*$ and $\frac{p_1^* - p_{1,r}^*}{p_{1,r}^*} < \frac{p_2^* - p_{2,r}^*}{p_{2,r}^*}$).*

In Proposition 3, we further show that in all cases p_2 increases more than p_1 (or decreases less than p_1) after the upgrade introduction, in terms of both the absolute amount of change and the percentage amount of change. This is also consistent with what we observe in Figure 1. Proposition 3 therefore indicates that the difference between the two price levels, which provides a measure for the firm's price dispersion, becomes larger after the firm introduces premium product upgrades. The result is formally stated in Corollary 1 below.

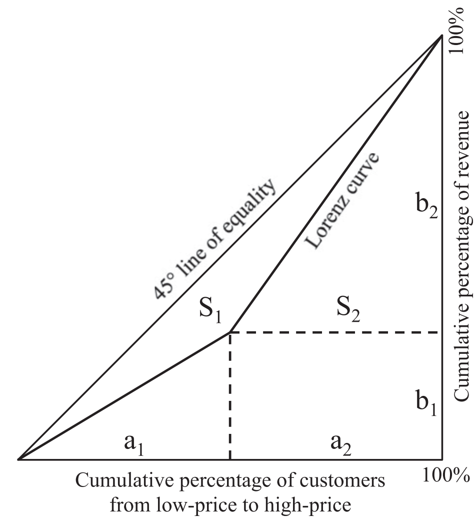
Corollary 1. *The difference between the two price levels increases after the upgrade introduction (i.e., $p_2^* - p_1^* > p_{2,r}^* - p_{1,r}^*$).*

It is worth mentioning that although the admission effect and the valuation effect work as competing forces regarding the change of each individual price level, they both result in an increase in the price dispersion. If the firm wants to attract more customers by decreasing regular product prices so that it could raise more revenues from selling upgrades, decreasing p_1 is more effective because p_1 directly determines how many customers in total purchase from the firm, whereas changing p_2 only affects what percentage of customers will pay the high price. Thus, the impact of the admission effect is stronger on p_1 than on p_2. Moreover, the valuation effect is stronger on p_2 than on p_1. Although the valuation effect has an upward impact on all prices, the upgrade-driven valuation enhancement is more significant for higher-valuation customers. Thus, the valuation effect results in p_2 increasing more than p_1 after the upgrade introduction. Therefore, both the admission and valuation effects result in an increase in the price dispersion of regular products after the firm introduces premium product upgrades.

Given our low-to-high pricing setting, the difference between the two price levels is one measure for the firm's price dispersion. However, the price dispersion among all passengers on the same plane could be determined by not only the price levels set by the airline but also the number of customers purchasing at each price level. We next consider the Gini coefficient as the measure for price dispersion. The Gini coefficient is a commonly used measure in economics for inequality among a population and is also our main measure for price dispersion in the empirical analysis. Mathematically, the Gini coefficient is the ratio of the area between the 45° line of equality and the Lorenz curve (which plots the proportion of the total income or consumption of the population that is cumulatively earned by the bottom x% of the population) to the total area below the 45° line of equality. The Gini coefficient ranges between 0 and 1. A Gini coefficient of 0 indicates perfect equality, whereby every individual has the exact same income or consumption. A Gini coefficient of 1 indicates maximal inequality. For a large population, if only one person has all the income or consumption and all others have none, the Gini coefficient will be very close to 1.

Figure 2 illustrates the Gini coefficient in our setting. a_1 and a_2 represent the percentages of customers paying price p_1 and p_2, respectively. b_1 and b_2 represent the percentages of revenue from customers paying price p_1 and p_2, respectively. S_1 and S_2 represent the areas above and below the Lorenz curve, respectively, which can be calculated from a_1, a_2, b_1, and b_2. The Gini coefficient is $\frac{S_1}{S_1+S_2}$. Let $G(p_1, p_2)$ and $G_r(p_1, p_2)$ denote the

Figure 2. Graphic Illustration of Gini Coefficient



Gini coefficients in the case with upgrades and the case without upgrades, respectively. Correspondingly, let G^* and G_r^* denote the Gini coefficients under the optimal pricing.

Proposition 4. *The Gini coefficient increases after the upgrade introduction (i.e., $G^* > G_r^*$).*

Proposition 4 states that the Gini coefficient becomes higher after the upgrade introduction. Thus, our theoretical finding regarding the change in price dispersion is robust if we account for the number of customers purchasing at different prices. We will provide a direct test of this hypothesis, highlighted as Hypothesis 2 below, in our empirical analysis.

Hypothesis 2. *Price dispersion increases after the firm introduces premium product upgrade.*

In summary, our theoretical findings generate the main hypotheses for our empirical analysis in Section 5. Our theoretical results predict that price dispersion should increase after the upgrade introduction (Hypothesis 2). We will further study the manner in which the increase in price dispersion is manifested. Note that the theoretical analysis suggests that the admission effect is stronger for p_1 than p_2, and the valuation effect is stronger for p_2 than p_1. However, whether each price is driven more by the admission effect or the valuation effect depends on the demand conditions. Therefore, we will empirically test the effect of upgrade introduction on individual price levels (Hypothesis 1a–1c).

4.2. Implications for Firm Revenue and Consumer Surplus

We now examine the effects of upgrade introduction on firm revenue and consumer surplus. First, consider the effect of upgrade introduction on the firm's revenue. Because the offering of upgrades provides the firm with another lever for price discrimination, the total revenue should always increase after the upgrade introduction. Proposition 5 confirms this result. Note that Proposition 5 holds (i.e., the revenue is strictly higher after the upgrade introduction) even when $w = 0$. Thus, the revenue improvement of upgrades is not simply a result of the fact that the presence of upgrades increases customers' valuations overall (e.g., through free upgrade opportunities) but is rooted in firm's ability to price-discriminate in a new dimension.

Proposition 5. *The firm's total revenue increases after the upgrade introduction (i.e.,* $\Pi^* > \Pi_r^*$*).*

Next, we study how the change in firm pricing due to the upgrade introduction affects consumer surplus. Let $C(p_1, p_2)$ and $C_r(p_1, p_2)$ denote the consumer surplus from purchasing regular products in the case with upgrades and the case without upgrades, respectively. Correspondingly, let C^* and C_r^* denote the consumer surplus under the optimal pricing.[9] We have

$$C_r^* = \int_{p_{1,r}^*}^{p_{2,r}^*} (v - p_{1,r}^*)\mathrm{d}v + \int_{p_{2,r}^*}^{1} (v - p_{2,r}^*)\mathrm{d}v,$$

and

$$C^* = \int_{p_1^*}^{p_2^*} (\hat{v} - p_1^*)\frac{1}{1+w}\mathrm{d}\hat{v} + \int_{p_2^*}^{1+w} (\hat{v} - p_2^*)\frac{1}{1+w}\mathrm{d}\hat{v}.$$

Proposition 6. *The consumer surplus increases after the upgrade introduction (i.e.,* $C^* > C_r^*$*).*

Proposition 6 states that under the new pricing strategy with upgrades, consumer surplus increases compared with the case without upgrades. As we have discussed in Section 4.1, price changes after the upgrade introduction are driven by two effects, the admission effect and the valuation effect. Owing to the sequential nature of the upgrades, the admission effect stems from the firm's incentive to decrease the prices and capture more customers. Because the result of the admission effect is toward pushing prices down and admitting more customers, this effect naturally increases consumer surplus. The valuation effect stems from the fact that customers' valuations are increased when upgrades are offered. Of course, the firm can also increase the price because of this. In our basic model, the increased "valuations" are shared between the firm and the customers so that customers have a net surplus.[10] Therefore, the upgrade strategy can increase consumer surplus and create a win–win situation, which is in contrast with previous literature finding for single-dimensional price discrimination (e.g., Leslie 2004).

4.3. Endogenous Upgrade Price

In our main model, we assume that the price for customers to purchase an upgrade from regular products to premium products is exogenous because this reflects the practice of the airline we worked with when they first introduced premium economy seating and matches our empirical setting. We now explore whether our results would be affected if the airline set the upgrade price in response to demand and coordinated the optimization of product prices and upgrade prices. Whether a customer purchases an upgrade will now depend on the upgrade price. Additionally, customers with different valuations for regular product may differ in their probabilities of purchasing upgrades. Therefore, we assume that the probability that a customer with original valuation v for regular products purchases a paid upgrade is $[u + \rho(v - \frac{1}{2}) - p_u]^+$, for any upgrade price p_u chosen by the firm. Note that the upgrade probability in the main model was $u + \rho(v - \frac{1}{2})$. Thus, the change reflects the idea that the probability of purchasing an upgrade to a premium seat decreases in the upgrade price p_u.

The firm's optimization problem is formulated as follows:

$$\begin{aligned}\max\ \Pi(p_1, p_2, p_u) &= p_2\frac{1+w-p_2}{1+w} + p_1\frac{p_2-p_1}{1+w} \\ &\quad + p_u\int_{p_1}^{1+w}\left[u + \rho\left(\frac{\hat{v}}{1+w} - \frac{1}{2}\right) - p_u\right]^+ \\ &\quad \cdot\frac{1}{1+w}\mathrm{d}\hat{v} \\ &= \frac{1}{1+w}[p_2(1+w-p_2) \\ &\quad + p_1(p_2 - p_1)] \\ &\quad + p_u\int_{\frac{p_1}{1+w}}^{1}\left[u + \rho\left(v - \frac{1}{2}\right) - p_u\right]^+\mathrm{d}v\end{aligned} \tag{3}$$

$$\text{s.t. } 0 \le p_1 \le p_2 \le 1+w,\ 0 < p_u \le u + \frac{|\rho|}{2}.$$

Note that the optimization of p_u is restricted to $0 < p_u \le u + \frac{|\rho|}{2}$. It is easy to verify that the optimal upgrade price is always strictly positive. An upper bound of $u + \frac{|\rho|}{2}$ indicates that the upgrade price does not exceed the highest probability that any customer purchases a paid upgrade. If this is violated, none of the customers purchases an upgrade.

We again investigate how the firm's price dispersion and revenue, as well as consumer surplus, change after the firm introduces premium product upgrades. The results are shown in Propositions C.1–C.6 and Corollary C.1 in Online Appendix C. In particular, all results for price dispersion carry through and generate the same hypotheses for the empirical analysis. The model also predicts that when the upgrade price is endogenously determined, both the firm revenue and the consumer surplus increase after the upgrade introduction. Thus, the insights from this paper do not depend on whether the upgrade price is endogenous.

5. Empirical Analyses

We first detail our data and identification strategy. We then present results from regressions exploring the impact of the option to upgrade to premium economy seating on price dispersion within the coach cabin, and from regressions analyzing the 10th and 90th percentiles of the price distribution separately to gain additional insights regarding the source of change in price dispersion. We also study the moderating impact of competition on the influence of the premium economy seating introduction on price dispersion. Finally, we present several robustness checks.

5.1. Industry Terminology, Data, and Identification Strategy

In this industry, a *market* refers to all flights serving an undirected origin–destination pair. For example, all flights departing from JFK and arriving at LAX as well as all flights departing from LAX and arriving at JFK serve the JFK–LAX market. A *flight instance* is identified by a departure date, flight number, origin, and destination. For clarity of terminology, it is important to distinguish a flight instance from a flight. A *flight* refers to the collection of flight instances across time for the same flight number serving the same market. For example, American Airlines 293 from New York (JFK) to Los Angeles (LAX) on June 10, 2015 departing at 8 pm is a particular flight instance, whereas the flight refers to all flight instances of AA293 serving the JFK–LAX market.

The data reflect ticket purchases for all flight instances that departed during the period that spanned the 12 weeks before, the week of, and the 12 weeks after the airline's introduction of the premium economy seating upgrade option on domestic markets. The data report the price that the airline charged for each seat it sold net of any surcharges, or taxes and fees collected by the government.[11] It also reports ticket details such as booking date, fare class, ticketing carrier, departure and arrival airports, flight number, and departure date.

We use three metrics that are common to the literature in characterizing the price dispersion on each flight instance: the Gini coefficient, the coefficient of variation (CV), and the interquartile range (IQR) of prices. The Gini coefficient is the most commonly used metric to describe price dispersion in the airline industry. It can be interpreted as half of the expected absolute difference between any two randomly drawn tickets. For example, a Gini coefficient of 0.25 on a plane implies an expected absolute price difference of 50% of the mean fare. The CV is the ratio of the standard deviation to the mean of the price distribution, a common metric used in operations management to describe variability. The IQR is the difference between the 75th and 25th percentiles of the price distribution. The main advantage of IQR is its robustness to outliers. We show that our findings are robust to the choice of different dispersion metrics. Moreover, we also investigate changes in the 10th and 90th percentiles of the price distribution, to shed light on the drivers of price dispersion movements after the introduction of the option to upgrade to a premium economy seat.

Table 1 documents the summary statistics of prices and price dispersion metrics across the 105,519 flight instances in our study. The first row presents the summary statistics for the average net price across all the passengers on a given flight instance. The average price has a mean of \$130.5; however, there is substantial variation across flight instances, as presented by the standard deviation and the percentile metrics. We also present summary statistics for the 10th percentile price ($P10$) and 90th percentile price ($P90$) of each flight instance. The average 10th percentile price on a flight is \$62.5, and the average 90th percentile price is \$234.5. Again, we see a considerable variation in these metrics across flight instances. The bottom three rows of Table 1 present summary statistics of the three price dispersion metrics we study. The average of the Gini coefficient is 0.277, which corresponds to an expected fare difference of 55% of the mean fare for two randomly selected passengers on a given flight instance. On average, the CV is 0.573, and the IQR is \$81.5. Albeit measuring price dispersion at the flight instance level, rather than for each market across a quarter, our dispersion metrics fall well within the range of the previously reported statistics.[12] Note that there is a large amount of variation in all price dispersion metrics across flight instances. A substantial part of this variation is accounted by systematic differences in price dispersion across markets and the seasonality in price dispersion over time. Our empirical analysis will separate sources of systematic variations across markets and time, to identify the impact of the introduction of the option to upgrade to premium economy seats on price dispersion.

We aim to identify the causal impact of the introduction of premium economy seating upgrades on the price dispersion within the coach class. A simple comparison of the price dispersion before and after the introduction of the option to upgrade to premium economy seating cannot be separated from changes in market conditions in the same time period. For example, if the price dispersion usually declines in summer months, and the upgrade introduction coincides with the beginning of summer, one may erroneously infer that the introduction of the upgrades decreased price dispersion, when price dispersion would have declined regardless.

To address potential confounds arising from such contemporaneous and noncausal variations, we conduct a difference-in-differences analysis, which compares

Table 1. Summary Statistics of Prices and Price Dispersion Metrics

			Percentiles				
	Mean	Standard deviation	10th	25th	50th	75th	90th
Average price	130.5	50.8	75.9	95.4	121.8	157.0	195.5
*P*10	62.5	27.0	36.4	44.7	56.8	73.7	95.4
*P*90	234.5	109.1	114.4	154.9	214.9	293.3	390.2
Gini	0.277	0.090	0.164	0.224	0.279	0.337	0.388
CV	0.573	0.207	0.328	0.440	0.562	0.702	0.833
IQR	81.5	57.8	26.1	44.1	67.6	104.6	152.2

Note. Reflects data across 105,519 flight instances.

the changes in price dispersion in the flight instances within markets that received premium economy seating (i.e., treated markets) with the changes in price dispersion in those that did not receive premium economy seating (i.e., control markets). The difference-in-differences design aims to control for changes in the price dispersion that would have occurred even if premium economy seating was not introduced. The identifying assumption is that the changes in price distribution in the control and treated markets over time would be comparable had there not been the introduction of premium economy seating in the treated markets.

Our ability to find proper control markets stems from exogenous delays in the airline's initial rollout plans. Even though the airline intended to roll out premium economy seating in all domestic markets at the same time, it failed to do so owing to manufacturer delays that were out of the airline's control. The airline had to work with several manufacturers to change the seating configuration on its aircrafts. One manufacturer of smaller-sized aircrafts had to provide an updated model to accommodate more leg space between the front rows of seats. Because of delays in aircraft availability with premium economy seating, the airline was able to introduce the premium economy seating upgrades in only approximately 70% of domestic markets at the initial rollout date. Importantly, all remaining markets received the premium economy seating within a few months after the initial rollout date as the updated aircrafts became available.[13]

The airline was mostly using the aircrafts from the delay-causing manufacturer to conduct frequent services between markets consisting of a medium-sized city on the one end (e.g., Cleveland, OH) and a hub on the other, as can be expected from a hub-and-spoke airline. For every market that did not receive the aircrafts with the premium economy seating at the time of the initial rollout date, we look for other markets that have the *same* medium-sized city but a different hub and that have received aircrafts with the premium economy seating. The markets that experienced delays in the introduction of premium economy seating serve as control markets for the markets that received the premium economy seating. In this manner, we create market groups consisting of treated (i.e., received upgrade option on time) and control (i.e., experienced delay) markets that share the same medium-sized city but have different hubs on the other end. We keep the medium-sized city constant so that the time-varying unobservables in the control markets, such as demand-side factors that may impact price distributions over time, are as similar as possible to those in the treated markets. For each market that received the premium economy seating at the initial rollout, we look for at least one market that did not receive it until after our data period. This exercise gives us 23 groups consisting of 78 markets, 3,753 flights, and 105,519 flight instances in the 25-week period of our study.

Clearly, treated and control markets may differ in their demand potential, because the airline decides which of the markets that needed the upgraded aircrafts should receive them as the aircrafts became available and may prioritize markets that carry more passengers or account for higher profits. We provide a comparison between treated and control markets in Table 2. As we expected, the control markets tend to be smaller than the treated markets, as reflected by the lower number of total passenger count and fewer carriers in the control markets. However, the airline's market share is similar across the treated and control markets, and the treated and control markets are also similar in terms of price dispersion.

Given the differences across control and treated markets in terms of market size, it is important to note that our identification approach does not require that the control and treated markets are similar in their market conditions. In the estimations that follow, we allow for the possibility that the flight instances in markets that received the upgraded aircrafts on time may be systematically different than those in markets that did not receive the upgraded aircrafts on time by including flight fixed effects. Flight fixed effects give us a strict control over any time-invariant cross-sectional unobservables, whether these unobservables reflect systematic differences across markets or across flights

Table 2. A Comparison of Treated and Control Markets

	Treated markets $n = 45$		Control markets $n = 33$		
	Mean	Standard deviation	Mean	Standard deviation	p-value
Average price	146.6	34.4	137.2	39.5	0.591
*P*10	70.9	19.7	65.4	21.5	0.590
*P*90	258.9	61.5	254.8	81.6	0.810
Gini	0.276	0.051	0.285	0.060	0.372
CV	0.563	0.124	0.594	0.123	0.158
IQR	89.0	27.7	89.8	40.8	0.876
Total passenger count	129,328	155,700	41,238	70,063	0.081
HHI	0.846	0.206	0.936	0.151	0.253
Airline's market share	0.876	0.183	0.883	0.288	0.722
Average competitor price	145.3	28.0	164.6	39.8	0.025
No. of carriers	2.689	1.164	1.545	0.938	0.006
Booklag (d)	34.4	5.99	37.5	17.6	0.421
Flight time (min)	124	51	104	51	0.791

Notes. This table reports the summary statistics of prices and price dispersion metrics, as well as market characteristics, for the period before the introduction of premium economy seating upgrades. It also reports p-values from a regression that assesses the difference in these covariates between the treated and control markets within each group. The data on market characteristics is obtained from the DB1B database for the two quarters that our data period spans. The total passenger count is the average number of passengers traveling in a given market in a given quarter.

within a market. As a result, if there are omitted factors that vary across markets and influence both the price dispersion and the airline's decision regarding which markets to assign the upgraded aircrafts to, those factors are accounted for by flight fixed effects.

In sum, our identification approach relies on examining difference-in-differences: we compare the change in the coach cabin price dispersion before and after the premium economy seating upgrade option is introduced in a treated market with the change in the coach cabin price dispersion in a control market over the same time period. The identifying assumption is that the changes in price dispersion over time, after controlling for cross-sectional systematic differences across markets, are comparable across treated and control markets in the same group. Importantly, this approach does not require the treated and control markets to have similar baseline market conditions or price dispersion rates. It suffices that the timing of the upgrade introduction in treated markets was exogenous and uncorrelated with other factors that would create a difference in price dispersion across treated and control markets. This assumption is often referred to as the parallel trends assumption. We offer supporting evidence in its favor in Section 5.5.

5.2. Price Dispersion Regressions

We estimate a series of weighted regressions in which the dependent variable is one of the price dispersion metrics and the weights are proportional to the average number of tickets sold on a flight. We exploit the panel dimension of our data to assess the impact of upgrade introduction on price dispersion while controlling for time-invariant flight-specific factors, as well as group-level seasonality, as follows:

$$DISP_{ifmgt} = \beta \cdot Upgraded_{mt} + \xi_f + \gamma_{gt} + \nu_{ifmgt}, \tag{4}$$

where i indexes the flight instance, f indexes the flight, m refers to the market, g refers to the group of treated and control markets that share the same medium-sized city, and finally t indexes the week. The upgrade introduction event at the market level is captured by the $Upgraded_{mt}$ variable, which equals 1 after a treated market receives the upgrade option, and 0 otherwise. Price dispersion, $DISP_{ifmgt}$, is measured in three different ways: the log-odds ratio of the Gini coefficient, the natural log of the CV, and the natural log of the IQR of prices, each estimated in separate regressions.[14] Flight fixed effects, ξ_f, control for time-invariant differences in price dispersion across flights. Such variation may reflect price discrimination or cost differences across different departure times in the same market, or systematic differences across markets. We also control for contemporaneous unobservables that are common to the markets in the same group through a full set of group-week dummies, γ_{gt}. These controls allow for a higher flexibility in seasonality specification than imposing a common seasonality across all markets, because they permit different seasonality patterns for each group.

We also recognize two potential sources of heteroskedasticity. First, flight instances on smaller flights tend to have fewer tickets, leading to noisier measures of price dispersion metrics. Therefore, we use a weighted generalized least squares approach in estimating the regression displayed in Equation (4), using weights proportional to the average number of tickets on a flight to account for different levels of noise across observations. Second, pricing decisions across different instances of the same flight may be correlated. Therefore, we cluster the standard errors at the flight level.

5.3. Price Percentile Regressions

The Gini coefficient is an attractive statistic to analyze because of the ability to directly glean insights of how price dispersion changes from a single dependent variable, and much of the literature has relied on analyzing this statistic. However, an increase in price dispersion may occur for several reasons. Our theoretical model leaves the exact mechanism as an empirical question: the price dispersion may increase because the lower end of the price distribution decreases more than the higher end, or because the higher end of the price distribution increases more than the lower end, or because the lower end decreases while the higher end increases. Price percentile regressions allow us to take a deeper look into what type of price changes—at the lower or the higher end of the price distribution—results in the change in price dispersion picked up by the price dispersion metrics.

To this end, we estimate the following regression:

$$\log P(k)_{ifmgt} = \beta \cdot Upgraded_{mt} + \xi_f + \gamma_{gt} + \nu_{ifmgt}, \quad (5)$$

where k is either the 10th or 90th percentile. Again, we weight the observations proportional to the average number of tickets sold on a flight and cluster the standard errors at the flight level.

5.4. Results

The results from the price dispersion and price percentile regressions are presented in Table 3. Columns (1)–(3) report the β coefficient estimates from Equation (4). The estimates indicate that the introduction of the upgrade option is associated with an increase in the Gini log-odds ratio by 0.180, the CV by 15%, and the IQR by 15%. Table 3 also reports the standard errors in parentheses below the point estimates and indicates the significance level of the estimates. The effect of the introduction of the upgrade option on price dispersion is significant at the 1% significance level across all dispersion metrics. In sum, the empirical results confirm our main theoretical prediction.

A look at β coefficient estimates from the percentile regressions in Equation (5), reported in columns (4) and (5) of Table 3, sheds further light on the manner in which price dispersion changes with the introduction of the upgrade option. The estimates show that the increase in price dispersion is driven both by a decrease in the lower end of the price distribution and an increase in the higher end of the price distribution. In particular, the upgrade introduction is associated with approximately a 3.5% decrease in the 10th percentile price, and approximately a 6% increase in the 90th percentile price.

Our theory model suggests that the airline may decrease ticket prices to attract more customers to purchase the upgrades, an effect we refer to as the admission effect. Presumably, the higher competitive pressure the airline faces, the more it needs to decrease its prices to attract customers away from purchasing with alternative carriers, especially at the low end of the price distribution. Conversely, if the airline has a large market share and commands a strong market power, decreasing its low-end prices will not yield a large increase in the demand from its already captive audience. Therefore, we hypothesize that the admission effect on the low-end prices will be stronger in markets where the airline faces more competition and weaker in markets where the airline has more market power. As a result, we expect to see a larger increase in price dispersion in response to the introduction of premium economy seating upgrades in markets where the airline faces more competition and a smaller increase in markets where it has more market power.

To test this hypothesis, we estimate

$$y_{ifmgt} = \beta \cdot Upgraded_{mt} + \theta \cdot Competition_m \times Upgraded_{mt} + \xi_f + \gamma_{gt} + \nu_{ifmgt},$$

where y_{ifmgt} refers to one of the price dispersion metrics or price percentiles we consider, and $Competition_m$ is a measure of the degree of competition in a market. We consider three metrics for $Competition_m$: the airline's

Table 3. Estimates from Our Main Specification

	$Gini^{logodd}$ (1)	$\log CV$ (2)	$\log IQR$ (3)	$\log P10$ (4)	$\log P90$ (5)
Upgraded	0.180***	0.148***	0.149***	−0.035***	0.060***
	(0.044)	(0.039)	(0.035)	(0.008)	(0.019)
Observations	105,519	105,519	104,363	105,519	105,519
Controls	Group × week fixed effects, flight fixed effects				
Adjusted R^2	0.3302	0.3046	0.2698	0.5800	0.5136

Note. The standard errors are clustered at the flight level.
***$p < 0.01$.

market share, the market's Herfindahl-Hirschman Index (HHI), and the average competitor price in the market. The HHI is the sum of the squares of the market shares of the airlines within the market and therefore reflects market concentration. By construction, HHI varies between 0 and 1, 1 representing a monopoly. Because our airline is the leading carrier in most of the markets we consider, HHI essentially captures how much market power the airline has. Again, we weight the observations proportional to the average number of tickets sold on a flight and cluster the standard errors at the flight level.

We expect the decline in low-end prices and the increase in price dispersion to be higher in magnitude in markets with higher competitive pressures. Because the airline's market share, the market's HHI, and the average competitor price in the market are all inversely related to the competitive pressure, we expect β to be positive and θ to be negative in price dispersion regressions, whereas we expect β to be negative and θ to be positive in the 10th percentile price regression. The results are presented in Table 4.

In line with our expectations, we see that the increase in the Gini coefficient and the CV after the upgrade introduction in a market is lower if the airline has a higher market share, if its market power is more concentrated (as indicated by a higher HHI), or if it faces a higher competitor price. However, the IQR, which measures the distance between the 25th and 75th percentiles, does not show the same sensitivity, plausibly because it is not sensitive to the changes in the very low end of the price distribution. Indeed, when we examine the 10th percentile of prices, we see that the difference in the price dispersion response is driven by the difference in how much the low-end of the price distribution drops in response to the introduction of premium economy seating. We do not find an effect on high-end prices.[15] Taken together, these results bolster the intuition of the admission effect: the airline decreases low-end prices to attract more customers who may purchase upgrades to the premium economy seating. In markets where the airline faces more competition, decreasing the low-end prices helps to attract new customers more than in markets where it has a high market share already.

In addition, we show that travelers with higher valuations for late booking not only have a higher willingness to pay for upgrades but also have a higher probability of securing the upgrades with a discount or for free. To this end, we merge individual ticket transactions data with the individuals' upgrade purchase data (5.8 million observations). First, we show that customers who paid higher prices for regular economy seats are more likely to purchase upgrades. As Table A.1 (in Online Appendix A) shows, we find that customers who purchased the upgrade also

Table 4. Estimates from an Investigation of Role of Competition

	$Gini^{logodd}$			log*CV*			log*IQR*			log*P10*			log*P90*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Upgraded	0.522***	0.558***	0.342***	0.444***	0.474***	0.290***	0.268**	0.240	0.236**	−0.157***	−0.217***	−0.140***	0.064	0.034	0.032
	(0.129)	(0.156)	(0.110)	(0.114)	(0.137)	(0.091)	(0.124)	(0.147)	(0.107)	(0.037)	(0.047)	(0.039)	(0.052)	(0.062)	(0.056)
HHI × *Upgraded*	−0.401***			−0.347***			−0.139			0.143***			−0.004		
	(0.120)			(0.104)			(0.132)			(0.043)			(0.052)		
MrkShr × *Upgraded*		−0.430***			−0.371***			−0.103			0.206***			0.030	
		(0.147)			(0.126)			(0.154)			(0.053)			(0.063)	
CompetPrice × *Upgraded*			−0.058*			−0.051**			−0.031			0.037***			0.010
			(0.032)			(0.026)			(0.034)			(0.014)			(0.018)
Observations	105,519	105,519	105,519	105,519	105,519	105,519	104,363	104,363	104,363	105,519	105,519	105,519	105,519	105,519	105,519
Controls	Group × week fixed effects, flight fixed effects														
Adjusted R^2	0.3320	0.3315	0.3305	0.3066	0.3060	0.3049	0.2699	0.2698	0.2698	0.5804	0.5805	0.5802	0.5136	0.5136	0.5136

Note. The standard errors are clustered at the flight level.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

purchased tickets that were approximately 20% more expensive ($26 higher) compared with customers who did not purchase the upgrade, who on average paid $127 (column (1) of Table A.1). Even when we exclude observations in which the upgrade was free (column (2) of Table A.1), we find that customers who purchased the upgrade also purchased tickets that were 12% more expensive ($15 higher) compared with customers who did not purchase the upgrade. This finding supports the notion that the customers who have higher valuations for regular product also have higher valuations for the upgrade. Second, we provide empirical evidence that customers with higher valuations for the ticket are more likely to receive upgrades for free. Table A.2 (in Online Appendix A) reports the estimates from a regression of ticket prices paid by travelers who eventually sat in the premium economy seats onto an indicator of whether the traveler received the upgrade for free. We see that compared with the customers who purchased paid upgrades, those customer who received upgrades for free purchased tickets that were approximately 16% more expensive ($22 higher). In sum, these results support the notion that the increase in the higher-end prices may be driven by the valuation effect, as suggested by our theoretical model.

5.4.1. Summary of Results. Overall, the estimates provide evidence of a positive relationship between the availability of the upgrade option and the price dispersion. Furthermore, the results show that the increase in the price dispersion is driven by both the decline in low-end prices and the increase in high-end prices. In addition, we see that the decline in low-end prices, and therefore the increase in price dispersion, is more pronounced in markets where the airline faces greater competition. Next, we check the sensitivity of our findings to several assumptions.

5.5. Robustness Checks

We have conducted a series of robustness checks. The resulting tables for robustness checks are provided in Online Appendix A.

5.5.1. Falsification Test. For our identification strategy to be valid, it has to be the case that the observed differences in the price dispersion between the treated and control markets after the introduction of the upgrade option would not have been observed had the upgrade option not been introduced. This assumption is often referred to as the parallel trends assumption. It can be violated if the timing of the upgrade introduction coincides with a pricing regime change that only impacted the treated markets. Given that we control for group-level seasonality in a flexible manner, we think that a coinciding demand- or cost-driven shock is unlikely to explain our results. However, we also recognize that the airline's market prioritization (i.e., which markets to introduce the premium economy seating first) may be correlated with trends in the market. For example, the identifying assumption would be violated if the airline prioritized the markets where the price dispersion was already increasing.

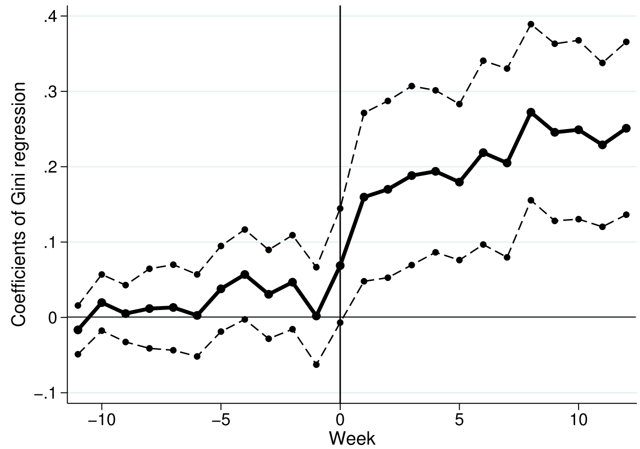
To allay such concerns, we conduct a falsification test that achieves two objectives. First, and most importantly, it tests whether the difference between treated and control markets showed a trend before the introduction of premium economy seating upgrades. Second, it allows us to see whether the difference increased immediately as a response to the treatment and whether the treatment had an increasing effect after its rollout. In particular, we estimate

$$y_{ifmgt} = \sum_{j=-11}^{12} \beta_j \cdot 1(t=j) \cdot Treated_m + \xi_f + \gamma_{gt} + \nu_{ifmgt},$$

where j indexes leads or lags from the introduction of premium economy seating upgrades. For example, $j = 0$ indicates the week during which the premium economy seating was introduced in treated markets, $j = -10$ indicates 10 weeks before the introduction, $j = 5$ indicates 5 weeks after the introduction. For the entire time period, $Treated_m$ equals 1 for treated markets and 0 for control markets.[16] Thus, β_j coefficients reflect the difference between the treated markets ($Treated_m = 1$) and the control markets ($Treated_m = 0$) within the same group across each time period $t = j$ as compared with the baseline time period, which we picked, without loss of generality, to be $t = -12$.

We report our results in Table A.3. Figure 3 plots β_j from the regression of the log odds of the Gini coefficient. We see no difference between the treated and control market pairs in each group before the

Figure 3. Coefficients β_j from the Regression of the log Odds of the Gini Coefficient



Note. The solid line shows β_j, the dashed lines show the 95% confidence interval of β_j, week 0 is the week of upgrade introduction.

introduction of premium economy seating upgrades. However, with the upgrade introduction at $t = 0$, the difference between control and treated markets in the same group suddenly jumps to a positive level and continues to increase for several weeks. Importantly, the sharp increase in the coefficients at the time of upgrade introduction cannot be reconciled by any prior trend in price dispersion in the treated versus the control markets. Although the parallel trends assumption cannot be tested directly, the results from the falsification test offer strong support for its validity and indicate that our identification strategy is appropriate in this context (Angrist and Krueger 1999).

5.5.2. Ruling out Contamination from First-Class Price Reactions. In our analyses we have focused on the airline's coach class pricing strategy, treating price changes in the coach class as independent of potential price changes in the first class as a result of the upgrade introduction. Particularly, we have ignored the potential of customers who would have otherwise bought first-class tickets to switch to premium economy seating after the upgrade introduction. One may worry that the change in price dispersion in the coach class, especially the increase in higher-end prices, could also be driven by potential switching behavior between cabins. However, we argue that the premium economy seating is fundamentally different from first-class seating, and the switching behavior from one to the other is unlikely. We arrive at this conclusion because of two pieces of empirical evidence.

First, if the airline was considering potential switching between cabins, we would expect a differential price response to the upgrade introduction in the coach cabin of flights that had first-class cabins compared with those that did not. In particular, because first-class passengers have high valuations, if there is cross-cabin substitution, we would expect the effect to impact how the high-end prices, and hence the price dispersion, in the coach cabin respond to the upgrade introduction. Therefore, we examine whether the effect of the upgrade introduction on the coach cabin prices is different in flights with first-class cabins versus those without.[17] Table A.4 reports the results. The increase in the high-end prices in the coach cabin does not differ according to whether the flight offers a first-class cabin, in support of the notion that customers who purchase first-class seating do not switch to the coach cabin for premium economy seating upgrades. Moreover, the changes in the price dispersion metrics and the low-end prices also are nondistinguishable according to whether the flight offers a first-class cabin.

We also provide a second piece of empirical evidence on this issue. If customers who would otherwise purchase first-class seats switched to purchasing premium economy seats after the upgrade introduction, we should also see a change in the transaction prices of first-class tickets. We thus examine the transaction prices and price dispersion within the first-class cabin. Table A.5 reports the results. We do not see any movement in the first-class prices. These results indicate that premium economy seating is fundamentally different from first-class seating and that potential switching between the coach cabin and the first-class cabin is unlikely and therefore support our focus on the coach class.

5.5.3. Intertemporal Price Dispersion. We take another perspective of looking at price dispersion by explicitly capturing the intertemporal aspect. To empirically test whether the intertemporal price dispersion increased as a result of the upgrade introduction, we test whether the sensitivity of ticket prices to the number of days the ticket is booked in advance ("booklag") became higher after the upgrade introduction. If prices of tickets booked later increased and prices of tickets booked earlier decreased, we would expect the slope of prices with respect to booklag to be steeper after the upgrade introduction. Therefore, we estimate

$$Price_{zifmgt} = \alpha_{mt} \cdot Booklag_{zifmgt} + \xi_f + \gamma_{gt} + \nu_{zifmgt},$$

where z indexes the ticket, i indexes the flight instance, f indexes the flight, m refers to the market, g refers to the group of treated and control markets, and finally t indexes the week. Importantly, we allow α_{mt} to respond to the upgrade introduction, and this response to differ in treated versus control markets. We specify

$$\alpha_{mt} = \alpha_0 + \alpha_1 After_t + \alpha_2 Treated_m + \alpha_3 After_t \times Treated_m$$

in a difference-in-differences fashion. Because prices decline with days between booking and departure, we expect α_0 to be negative. We expect α_3 also to be negative, because our hypothesis is that the slope of prices with respect to booklag becomes steeper after the upgrade introduction. The estimates are reported in Table A.6. As hypothesized, we find α_0 and α_3 to be negative. The sensitivity of prices to advanced booking days increased by approximately 10% (i.e., α_3/α_0) after the upgrade introduction in treated markets compared with control markets.

5.5.4. Seasonality Specification. Table 3 presents our favored specification, which includes group-week dummies. Table A.7 presents results from a specification that features shared seasonality controls (i.e., week dummies) rather than group-specific seasonality controls. The results indicate that our main conclusions are not susceptible to assuming a common seasonality, but the results are more pronounced when we allow for flexible seasonality patterns across groups.

5.5.5. Standard Error Specification. Finally, we consider two alternative standard error specifications: (1) market-level clustering to account for potential across-flight within-market correlations, and (2) two-way clustering by flight and day to account for potential correlations across days caused by the holiday effect. Table A.8 presents the results from a specification that clusters the standard errors at the market, rather than flight, level. Table A.9 presents the results from a specification that two-way clusters the standard errors by flight and day. The results are still significant at similar levels and our conclusions remain unchanged.

6. Conclusion and Discussion

Across industries, there has been a proliferation of price discrimination strategies to extract multidimensionally heterogeneous customer valuations. For example, in response to the recognition that some travelers may value comfort more than others, several major U.S. airlines have recently introduced the option to purchase premium economy seats by paying an upgrade fee after purchasing regular economy seats. Although the traditional intertemporal price discrimination across economy seats exploits the customers' heterogeneous valuations for scheduling flexibility, this new upgrade option allows the airlines to also price-discriminate in a new valuation dimension of customers within the coach class—comfort. Similarly, hotels, cruise lines, and entertainment venues have also introduced vertical upgrades as a way to complement their intertemporal pricing policies. These industry practices motivate us to study the following research questions. When companies introduce a vertical upgrade, does it change the extent to which they price-discriminate on their base products? If so, how and why? How are customers in different segments affected by the introduction of the upgrades?

This paper takes a first step in answering these questions in the context of the airline industry. We examine how introducing the option to upgrade to premium economy seating affected the airlines' price dispersion across the economy seats, through both theoretical and empirical analyses. To the best of our knowledge, this paper is among the first to shed an empirical light on the interaction between different price discrimination levers in a multidimensional price discrimination context. We find that price dispersion increases after the upgrade introduction, owing to both increased valuations of customers at the high end and the firms' incentive to lower prices to admit more customers at the low end. Particularly, lower-end prices declined by 3.5% and higher-end prices increased by 6% after the upgrade introduction.

How does the introduction of upgrades, and the resulting increase in price dispersion in the coach cabin, impact the firm and its consumers? Recall that the theoretical analysis predicts the firm's total revenue to increase with the upgrade introduction. In our empirical application, the increase in total revenue from coach class was about \$850 on average (Table A.10, Online Appendix A). Compared with the average revenue of approximately \$26,500 per flight instance, the total revenue increase is on the order of 4%, which in a low-margin industry is highly welcomed. Moreover, recall that the theoretical analysis suggests consumer surplus may increase for two reasons. First, because the airline decreases prices to leisure travelers, those travelers are unambiguously better off. Second, the upgrade introduction may increase the valuations for the base product for frequent fliers who anticipate a high chance of free upgrades. Although our empirical analysis does not afford a direct test of consumer surplus, we present empirical evidence consistent with the above two reasons for an increased consumer surplus. There, our result suggests that the upgrade strategy can create a win–win situation for both the firm and its consumers.

We acknowledge two limitations of our empirical analysis. First, the results are obtained from a selected group of markets for which we could find appropriate control markets. In particular, we caution the reader that the effect of the upgrade option on price dispersion is mainly informed by routes serving medium-size cities. Second, the empirical application is based on one major U.S. airline. Because of its large hub-and-spoke network, sophisticated revenue management systems, and extensive marketing abilities, this airline was able to introduce premium economy seating nationally and inform its customers about it in an extensive marketing campaign. Other smaller or more regional airlines may have different experiences if their customers are not well informed or if their revenue management systems are not as sophisticated. We believe, however, that the results in this paper will help guide their experiences.

In sum, this paper provides insights to firms regarding how to adjust price discrimination strategies (i.e., revenue management strategies) for the base product when a new price discrimination lever, in the form of an add-on product, is introduced. To maximize revenue, it is important to understand how the firm's main product revenue management strategy interacts with the add-on policy and how this interaction differentiates across customer segments (e.g., business travelers versus leisure travelers for travel firms). We hope that our findings not only provide insights for firms considering the introduction of vertically differentiated products as add-ons and inform regulators about the potential effects of multidimensional price discrimination endeavors of firms, but also inspire future work that combines analytical and empirical investigations to study managerial considerations in using multidimensional price discrimination policies across various industries.

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Endnotes

[1] The usage of premium seating upgrades as an additional perk to elite travelers has been recognized along with its value as a price discrimination tool (McCartney 2015).

[2] The data were provided on the condition of complete anonymity of the airline's identity.

[3] DB1B is an external and public database maintained by the Bureau of Transportation Statistics (http://www.transtats.bts.gov/).

[4] At the time of premium economy seating introduction, the airline we worked with offered the premium economy seats only as an add-on product that was available after the purchase of regular economy seats.

[5] Under this setting, the optimal prices are increasing in time, and forward-looking behavior of customers is irrelevant.

[6] At the time the airline we worked with introduced premium economy seating, the department that set the upgrade price was not coordinating with the revenue management team. The upgrade price was set mainly as a function of the distance of the trip, rather than in response to differences in demand and market conditions for a given route.

[7] See Zhang et al. (2018) for another study that examines customers' responses to the firm's price discrimination strategy.

[8] Our results are robust even if customers exhibit reference-dependent upgrade purchase behavior (i.e., if a customer already paid a higher price for the regular product, she may be more likely to purchase an upgrade).

[9] Note that although we do not explicitly model the consumer surplus from purchasing upgrades, $C^* > C_r^*$ (Proposition 6) implies that the total consumer surplus (including the consumer surplus from purchasing upgrades) increases after the upgrade introduction, because customers will only purchase the upgrades if they receive a nonnegative surplus from purchasing the upgrades.

[10] We note that our basic model has no capacity constraints. When we consider a model with capacity constraints (Online Appendix B), consumer surplus still increases after the upgrade introduction as long as the total capacity is not too constrained. However, if the capacity is very restricting, consumer surplus could decrease.

[11] Karlsson et al. (2004) estimate that the effective tax rate on the average base fare was 10.9% in 1993 and 15.5% in 2002. Within the contiguous 50 states, these charges include the U.S. domestic transportation tax (7.5%), U.S. federal flight segment fee ($3.80), September 11 security fee (up to $5.00, $2.50 per segment), and passenger facility charge (up to $4.50). There are no significant changes to these taxes and fees in the period our data span.

[12] Gerardi and Shapiro (2009) examine ticket transactions between 1993 and 2006 and report a Gini coefficient corresponding to an expected fare difference of 44%. Evans and Kessides (1993) report CV = 0.336 for economy class tickets between 1984 and 1988, and Hernandez and Wiggins (2014) report CV = 1.02 for the year 2004 over all service classes.

[13] To maintain anonymity of the airline, we cannot reveal the exact timing of the introduction of premium economy seating, the aircraft manufacturer that caused delays, or other related details.

[14] Because the Gini coefficient is bounded between 0 and 1, we use the Gini log-odds ratio in our regressions, which is given by $Gini^{logodd} = \ln[Gini/(1 - Gini)]$ and produces an unbounded statistic. We also follow the previous literature on pricing regressions and use natural logs of CV and IQR. This approach has two advantages: it decreases the sensitivity of the results to outliers and allows us to interpret the results in percentage change terms. Our results are not sensitive to these transformations.

[15] The reason why the airline may not have needed to decrease higher-end prices by a larger degree in more-competitive markets is the following. The customers that pay higher prices may be those who hold high status with the airline. These customers receive the upgrades at highly discounted prices or for free, thus their upgrade-driven valuation enhancement is much stronger. Therefore, the airline may not want or need to decrease ticket prices for these customers when it faces greater competition.

[16] Note the difference between the *Treated* variable in this regression and the *Upgraded* variable in Equation (4). Both variables equal 0 for all periods for the control markets. However, the *Treated* variable equals 1 for the treated market for the entire period, whereas the *Upgraded* variable in Equation (4) equals 0 before the upgrade introduction, and 1 after.

[17] We define a flight to have a first-class cabin if at least one first-class ticket was sold on any of its flight instances. Under this definition, 1,574 flights (accounting for 42% of all flights) are classified as flights that have first-class cabins. Our results are robust if we alternatively define a flight as having a first-class cabin if the percentage of flight instances for which a first-class ticket was sold is greater than 10%, 50%, or 90%.

References

Angrist JD, Krueger AB (1999) Empirical strategies in labor economics. *Handbook of Labor Economics*, vol. 3 (Elsevier, Amsterdam), 1277–1366.

Armstrong M (1996) Multiproduct nonlinear pricing. *Econometrica* 64(1):51–75.

Armstrong M (1999) Price discrimination by a many-product firm. *Rev. Econom. Stud.* 66(1):151–168.

Armstrong M, Rochet JC (1999) Multi-dimensional screening: A user's guide. *Eur. Econom. Rev.* 43(4):959–979.

Atkinson SE, Ramdas K, Williams JW (2016) Robust scheduling practices in the US airline industry: Costs, returns, and inefficiencies. *Management Sci.* 62(11):3372–3391.

Aviv Y, Pazgal A (2008) Optimal pricing of seasonal products in the presence of forward-looking consumers. *Manufacturing Service Oper. Management* 10(3):339–359.

Aydin G, Ziya S (2008) Pricing promotional products under upselling. *Manufacturing Service Oper. Management* 10(3):360–376.

Besbes O, Zeevi A (2009) Dynamic pricing without knowing the demand function: Risk bounds and near-optimal algorithms. *Oper. Res.* 57(6):1407–1420.

Bitran G, Caldentey R (2003) An overview of pricing models for revenue management. *Manufacturing Service Oper. Management* 5(3):203–229.

Borenstein S, Rose NL (1994) Competition and price dispersion in the U.S. airline industry. *J. Political Econom.* 102(4):653–683.

Broder J, Rusmevichientong P (2012) Dynamic pricing under a general parametric choice model. *Oper. Res.* 60(4):965–980.

Çakanyıldırım M, Özer Ö, Zhang X (2017) Dynamic pricing and timing of upgrades. Working paper, University of Texas, Dallas.

Chellappa RK, Sin RG, Siddarth S (2011) Price formats as a source of price dispersion: A study of online and offline prices in the domestic U.S. airline markets. *Inform. Systems Res.* 22(1):83–98.

Chen Q, Duenyas I, Jasin S (2016) Real-time dynamic pricing with minimal and flexible price adjustment. *Management Sci.* 62(8):2437–2455.

Chen Q, Jasin S, Duenyas I (2018) A nonparametric self-adjusting control for multi-product pricing with limited resources. *Math. Oper. Res.* Forthcoming.

Cho M, Fan M, Zhou YP (2009) Strategic consumer response to dynamic pricing of perishable products. *Consumer-Driven Demand and Operations Management Models* (Springer, New York), 435–458.

Chun SY, Iancu DA, Trichakis N (2017) Loyalty program liabilities and point values. Working paper, Georgetown University, Washington, DC.

Chun SY, Ovchinnikov A (2017) Strategic consumers, revenue management, and the design of loyalty programs. Working paper, Georgetown University, Washington, DC.

Clemons EK, Hann IH, Hitt LM (2002) Price dispersion and differentiation in online travel: An empirical investigation. *Management Sci.* 48(4):534–549.

Cornia M, Gerardi KS, Shapiro AH (2012) Price dispersion over the business cycle: Evidence from the airline industry. *J. Indust. Econom.* 60(3):347–373.

Cui R, Shin H (2018) Sharing aggregate inventory information with customers: Strategic cross-selling and shortage reduction. *Management Sci.* 64(1):381–400.

Cui Y, Duenyas I, Sahin O (2018) Pricing of conditional upgrades in the presence of strategic consumers. *Management Sci.* 64(7):3208–3226.

Dai M, Liu Q, Serfes K (2014) Is the effect of competition on price dispersion nonmonotonic? Evidence from the U.S. airline industry. *Rev. Econom. Statist.* 96(1):161–170.

Deshpande V, Arıkan M (2012) The impact of airline flight schedules on flight delays. *Manufacturing Service Oper. Management* 14(3):423–440.

Elmaghraby W, Keskinocak P (2003) Dynamic pricing in the presence of inventory considerations: Research overview, current practices, and future directions. *Management Sci.* 49(10):1287–1309.

Etzioni O, Tuchinda R, Knoblock CA, Yates A (2003) To buy or not to buy: Mining airfare data to minimize ticket purchase price. *Proc. 9th ACM SIGKDD Internat. Conf. on Knowledge Discovery and Data Mining* (ACM, New York), 119–128.

Evans WN, Kessides IN (1993) Localized market power in the U.S. airline industry. *Rev. Econom. Statist.* 75(1):66–75.

Gaggero AA, Piga CA (2011) Airline market power and intertemporal price dispersion. *J. Indust. Econom.* 59(4):552–577.

Gallego G, Hu M (2014) Dynamic pricing of perishable assets under competition. *Management Sci.* 60(5):1241–1259.

Gallego G, Stefanescu C (2009) Upgrades, upsells and pricing in revenue management. Working paper, Columbia University, New York.

Gallego G, van Ryzin G (1994) Optimal dynamic pricing of inventories with stochastic demand over finite horizons. *Management Sci.* 40(8):999–1020.

Gallego G, van Ryzin G (1997) A multiproduct dynamic pricing problem and its applications to network yield management. *Oper. Res.* 45(1):24–41.

Gerardi KS, Shapiro AH (2009) Does competition reduce price dispersion? New evidence from the airline industry. *J. Political Econom.* 117(1):1–37.

Hayes KJ, Ross LB (1998) Is airline price dispersion the result of careful planning or competitive forces? *Rev. Indust. Organ.* 13(5):523–541.

Hernandez MA, Wiggins SN (2014) Nonlinear pricing strategies and competitive conditions in the airline industry. *Econom. Inquiry* 52(2):539–561.

Karlsson J, Odoni A, Yamanaka S (2004) The impact of infrastructure-related taxes and fees on domestic airline fares in the U.S. *J. Air Transport Management* 10(4):283–291.

Leslie P (2004) Price discrimination in broadway theater. *RAND J. Econom.* 35(3):520–541.

Levin Y, McGill J, Nediak M (2009) Dynamic pricing in the presence of strategic consumers and oligopolistic competition. *Management Sci.* 55(1):32–46.

Levin Y, McGill J, Nediak M (2010) Optimal dynamic pricing of perishable items by a monopolist facing strategic consumers. *Production Oper. Management* 19(1):40–60.

Li J, Granados N, Netessine S (2014) Are consumers strategic? Structural estimation from the air-travel industry. *Management Sci.* 60(9):2114–2137.

Li J, Netessine S (2011) Partnering with competitors–An empirical analysis of airline alliances and multimarket competition. Working paper, INSEAD, Fontainebleau, France.

Liu Q, Zhang D (2013) Dynamic pricing competition with strategic customers under vertical product differentiation. *Management Sci.* 59(1):84–101.

Maglaras C, Meissner J (2006) Dynamic pricing strategies for multiproduct revenue management problems. *Manufacturing Service Oper. Management* 8(2):136–148.

Mantin B, Koo B (2009) Dynamic price dispersion in airline markets. *Transportation Res. Part E: Logist. Transportation Rev.* 45(6): 1020–1029.

McAfee RP, McMillan J (1988) Multidimensional incentive compatibility and mechanism design. *J. Econom. Theory* 46(2):335–354.

McCartney S (2015) The growing perks of premium economy. *Wall Street Journal* (August 12), http://www.wsj.com/articles/the-growing-perks-of-premium-economy-1439397061.

Nicolae M, Arıkan M, Deshpande V, Ferguson M (2017) Do bags fly free? An empirical analysis of the operational implications of airline baggage fees. *Management Sci.* 63(10):3187–3206.

Ramdas K, Williams J, Lipson M (2013) Can financial markets inform operational improvement efforts? Evidence from the airline industry. *Manufacturing Service Oper. Management* 15(3): 405–422.

Rochet JC, Chone P (1998) Ironing, sweeping, and multidimensional screening. *Econometrica* 66(4):783–826.

Stamatopoulos I, Tzamos C (2018) Design and dynamic pricing of vertically differentiated inventories. Working paper, University of Texas, Austin.

Stavins J (2001) Price discrimination in the airline market: The effect of market concentration. *Rev. Econom. Statist.* 83(1):200–202.

Stellin S (2013) Finding deals with paid upgrades. *New York Times* (June 19), http://www.nytimes.com/2013/06/23/travel/finding-deals-with-paid-upgrades.html.

Su X (2007) Intertemporal pricing with strategic customer behavior. *Management Sci.* 53(5):726–741.

Talluri K, van Ryzin G (2005) *The Theory and Practice of Revenue Management* (Springer, New York).

Yılmaz Ö, Pekgün P, Ferguson M (2017) Would you like to upgrade to a premium room? Evaluating the benefit of offering standby upgrades. *Manufacturing Service Oper. Management* 19(1):1–18.

Zhang DJ, Dai H, Dong L, Qi F, Zhang N, Liu X, Liu Z, Yang J (2018) How do price promotions affect customer behavior on retailing platforms? Evidence from a large randomized experiment on alibaba. Working paper, Washington University, St. Louis.